

Simulation and Energy Integration of Crude and Vacuum Distillation Units using Aspen Plus

Group 5

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Abstract

This study presents the simulation and energy integration of a Crude Distillation Unit (CDU) and Vacuum Distillation Unit (VDU) using Aspen Plus V14. The work focuses on modeling petroleum fractionation using pseudo-components, analyzing product yields, and optimizing energy usage through pinch analysis. The integration of heat recovery systems significantly reduces external utility requirements, improving process efficiency.

1 Introduction

Crude oil refining involves separating hydrocarbons into fractions based on boiling point differences. The CDU performs primary separation, while the VDU processes heavier fractions under reduced pressure to prevent thermal degradation. Efficient operation of these units requires careful control of temperature, pressure, and internal heat integration.

2 Objectives

The objectives of this work are to simulate CDU and VDU systems, analyze the effect of operating conditions on product yields, and apply pinch analysis to optimize energy consumption.

3 Simulation Basis and Methodology

3.1 Software and Model Selection

Aspen Plus V14 was used with the PetroFrac model. PetroFrac is specifically designed for petroleum refining and enables accurate simulation of complex distillation systems involving pseudo-components, side draws, and pump-around circuits.

3.2 Property Method

The **Peng-Robinson equation of state** was selected due to its suitability for hydrocarbon systems. It accurately predicts vapor-liquid equilibrium behavior under high temperature and pressure conditions.

3.3 Components

The system includes light hydrocarbons and pseudo-components derived from crude assay:

Component	Description
H ₂ O	Water
C1	Methane
C2	Ethane
C3	Propane
iC4	Iso-butane
nC4	n-Butane
iC5	Iso-pentane
nC5	n-Pentane
Pseudo-components	Heavy fractions (TBP based)

3.4 Crude Assay

Crude oil characterization was performed using TBP curves, density, and volume fraction data to generate pseudo-components representing real petroleum fractions.

4 Detailed Process Description

4.1 Overall Flowsheet

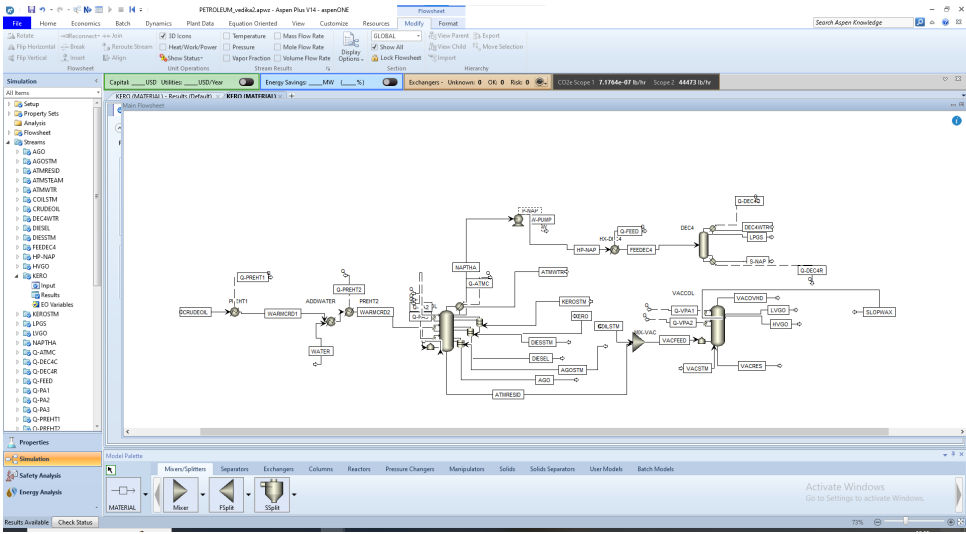


Figure 1: process flowsheet

4.2 Process Flow Diagram (PFD) Block Description

Block Name	Function	Engineering Justification
Crude Mixer (CRUDEOIL)	Blends crude feed streams	Ensures uniform feed composition. Blending is necessary because refinery feedstocks vary in composition and directly affect separation efficiency and product distribution.
Preheat Exchangers (PREHT1, PREHT2)	Heat crude using hot product streams	Reduces furnace duty by recovering heat from hot streams, thereby improving energy efficiency and minimizing external utility requirements.
Water Injection (WATER)	Injects water into crude stream	Helps remove salts and impurities, preventing fouling, corrosion, and catalyst poisoning in downstream equipment.
Furnace (PREHT2 outlet)	Raises crude temperature to 350–370°C	Provides the required heat for partial vaporization before entering the atmospheric column, enabling effective separation.
Atmospheric Column (ATMCOL)	Separates crude into fractions	Performs primary fractionation using vapor-liquid equilibrium across multiple stages, separating hydrocarbons based on boiling range.
Condenser (Q-ATMC)	Condenses overhead vapor	Generates reflux which enhances separation efficiency by improving vapor-liquid contact in upper stages.
Reboiler (Bottom of ATMCOL)	Supplies heat at column bottom	Generates upward vapor flow necessary to maintain vapor-liquid equilibrium and drive separation of heavier components.
Pump-Arounds (PA1, PA2)	Circulate liquid within column	Improve internal heat recovery, reduce condenser and reboiler duties, and maintain stable temperature profiles within the column.
Side Draws (NAPHTA, KERO, DIESEL, AGO)	Withdraw product streams	Enable separation of crude into commercially valuable products at different boiling ranges.
Vacuum Feed Mixer (MIX-VAC)	Mixes atmospheric residue streams	Prepares feed for vacuum distillation and ensures stable flow conditions under vacuum operation.
Vacuum Furnace (VACFEED heating)	Heats residue before VDU	Provides sufficient energy for vaporization of heavy hydrocarbons without causing thermal cracking.
Vacuum Column (VACCOL)	Separates heavy fractions under vacuum	Operates at low pressure to reduce boiling points and prevent thermal degradation of heavy hydrocarbons.
Vacuum Products (LVGO, HVGO, VACRES)	Final heavy product streams	Recover valuable heavy fractions such as gas oils and residue for further downstream processing or upgrading.

5 Simulation Results

5.1 Material Balance

Product	Flow (lb/hr)
Naphtha	312350
Diesel	312350
Kerosene	94740
HVGO	348651

6 Sensitivity Analysis

6.1 Effect of Furnace Temperature

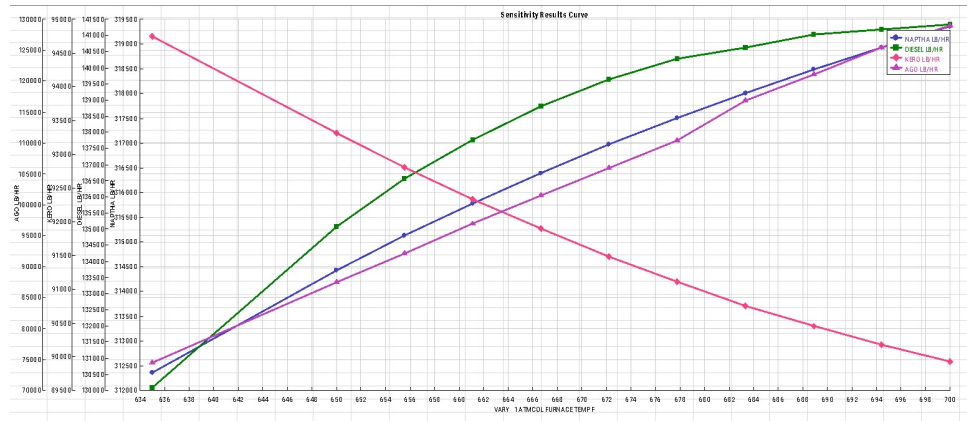


Figure 2: Effect of furnace temperature on product yields

- Increasing furnace temperature enhances vaporization of lighter hydrocarbons.
- Naphtha and diesel yields increase with temperature.
- Kerosene yield decreases as intermediate fractions shift toward lighter products.
- AGO yield shows an increasing trend due to improved separation of heavier fractions.
- Excessively high temperatures may lead to thermal cracking and energy inefficiency.

6.2 Effect of Pump-Around Flow Rate

Pump-around flow rate significantly affects internal heat distribution. Increasing pump-around enhances internal reflux and improves separation efficiency, but excessive flow may reduce vapor traffic and destabilize the column.

6.3 Effect of Steam Rate

Steam injection reduces hydrocarbon partial pressure and improves stripping efficiency. Higher steam rates enhance recovery of lighter components but increase utility consumption.

6.4 Effect of Feed Rate

Increasing feed rate increases throughput but may reduce separation efficiency due to tray overloading and reduced residence time.

7 Pinch Analysis

7.1 Objective

Pinch analysis was performed to minimize external heating and cooling requirements by maximizing internal heat recovery.

7.2 Pinch Point

The pinch point was identified at 180°F (hot) and 160°F (cold), corresponding to the minimum temperature approach.

7.3 Minimum Utilities

The minimum heating and cooling requirements were calculated as 326.86 MMBtu/hr and 518.48 MMBtu/hr respectively.

7.4 Composite Curves

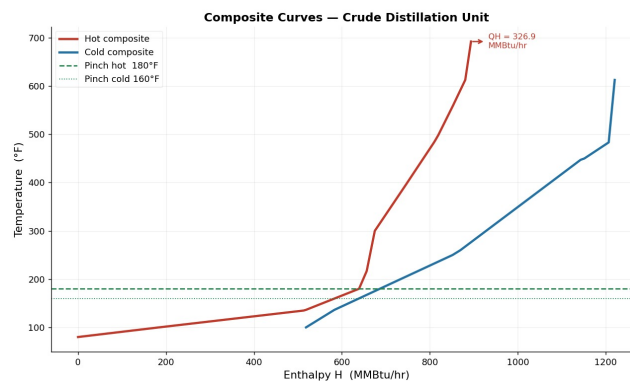


Figure 3: Composite curves

- The hot composite curve lies above the cold composite curve, indicating availability of recoverable heat within the process.
- The minimum vertical distance between the curves occurs at approximately 180°F (hot) and 160°F (cold), defining the pinch point.
- Above the pinch, the system has excess heat which can be internally recovered.
- Below the pinch, external heating is required to meet process demands.
- The horizontal gap at the top corresponds to the minimum heating requirement (Q_H).
- The gap at the lower end represents the minimum cooling requirement (Q_C).

7.5 Grand Composite Curve

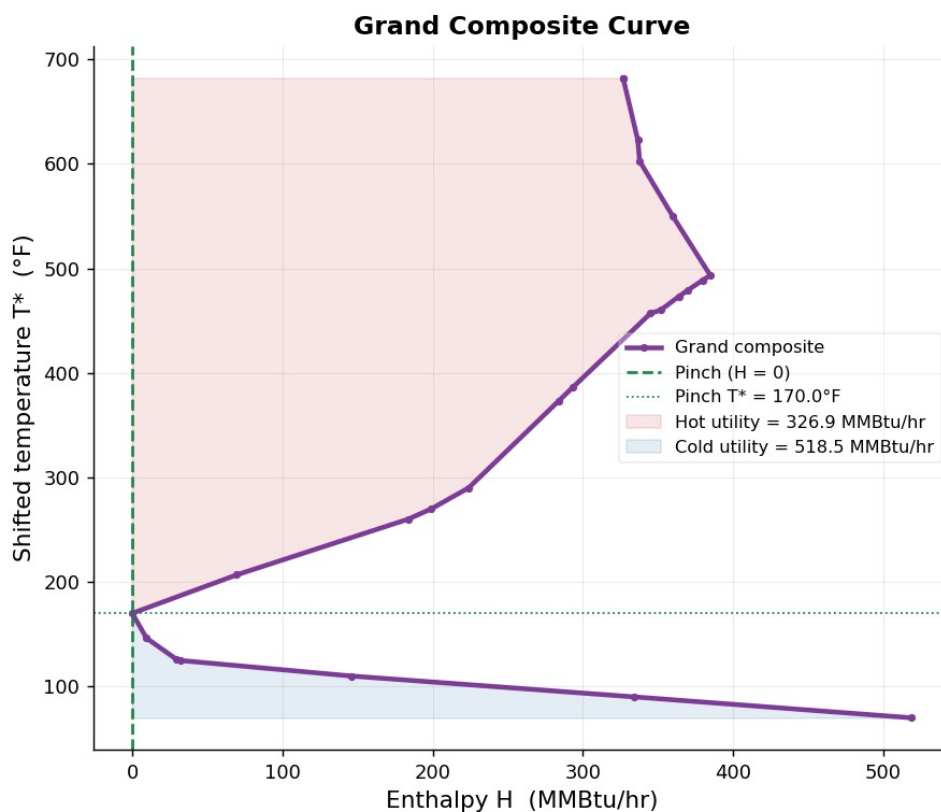


Figure 4: Grand composite curve

- The grand composite curve identifies temperature regions of heat surplus and deficit.
- Above the pinch, excess heat is available and can be utilized for preheating cold streams.

- Below the pinch, external heating is required to satisfy energy demand.
- The curve provides guidance for optimal placement of heat exchangers and utilities.
- The horizontal displacement at zero enthalpy represents the minimum hot utility requirement.

7.6 ΔT_{min} Sensitivity Analysis

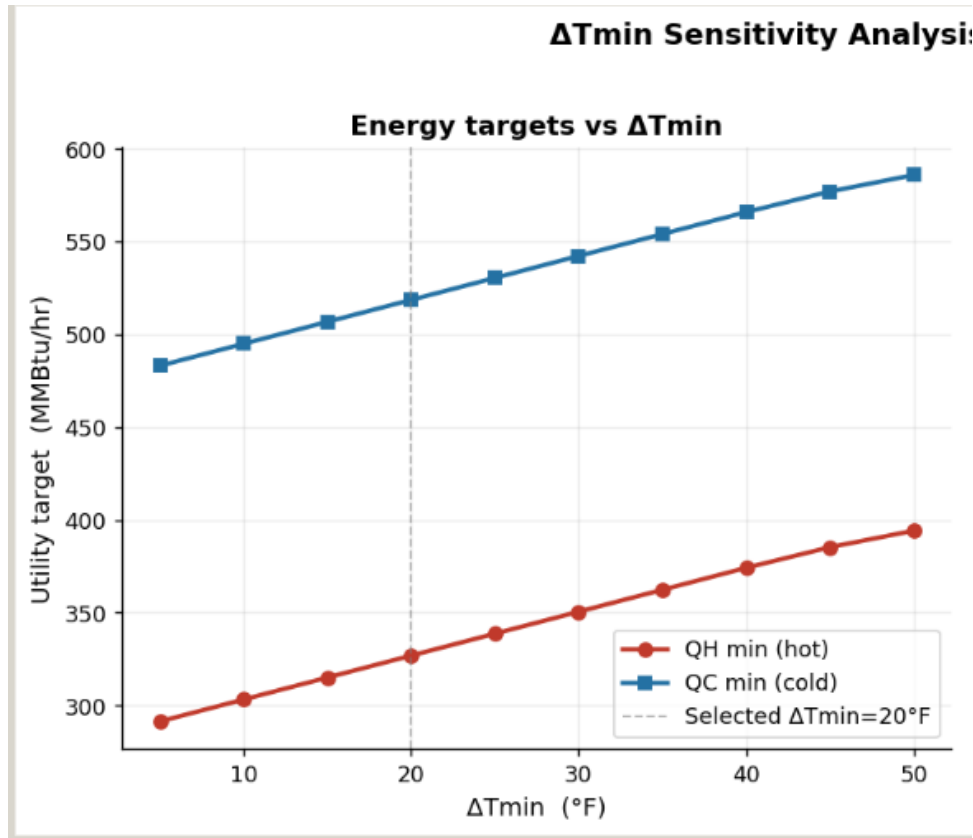


Figure 5: ΔT_{min} sensitivity analysis

- Increasing ΔT_{min} leads to an increase in both heating (Q_H) and cooling (Q_C) utility requirements.
- Lower ΔT_{min} values improve heat recovery but require larger heat exchanger area.
- Higher ΔT_{min} values reduce capital cost but increase operating cost.
- The selected ΔT_{min} of 20°F represents a compromise between energy efficiency and economic feasibility.
- The trend shows a nearly linear increase in utility demand with increasing ΔT_{min} .

8 Challenges Faced

Convergence issues were encountered due to improper stage selection, incorrect pump-around placement, and over-specification of variables. These were resolved by adjusting feed stage location, reducing constraints, and stabilizing heat duties.

9 Conclusion

The simulation successfully demonstrates the importance of proper configuration of CDU and VDU systems. Pump-arounds and heat integration significantly improve efficiency. Pinch analysis provides valuable insights into energy optimization and utility reduction.

10 Future Work

Future work includes detailed heat exchanger network design, dynamic simulation, and optimization of refinery operations.

11 References

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12 Author Contributions

All authors contributed equally to this work. Each member was involved in the simulation development, data analysis, interpretation of results, and preparation of the final report.